

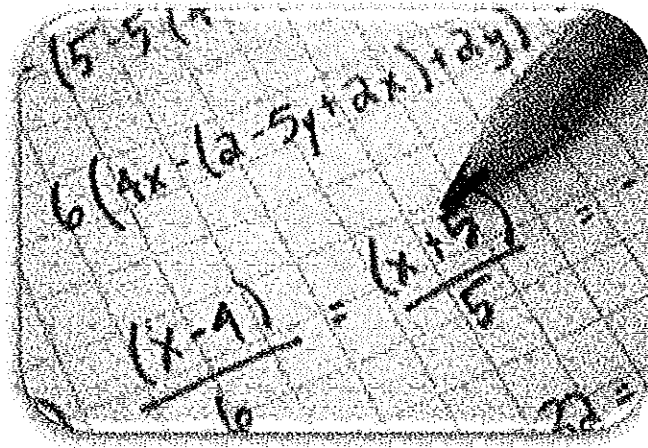
Full Name: SOLUTIONS



MATHEMATICS YEAR 8

Investigation – Optimisation of Area

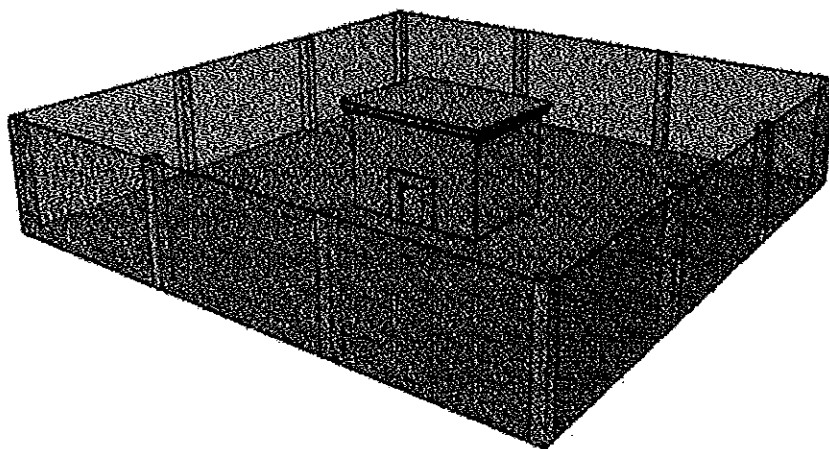
Semester 2 2015



Information for Students

Working time:	45 minutes
Marks available:	52 marks

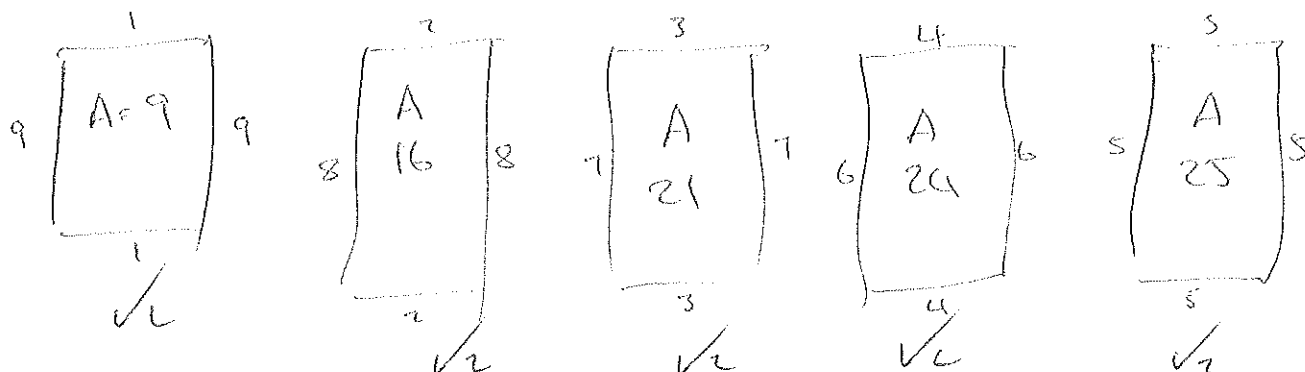
You have been given the task of designing a series of chook runs. In each situation you will be provided with a different length of chicken wire with which to construct a rectangular run.



Situation One (13 marks)

You have 20 metres of chicken wire and you need to enclose a rectangular run.

Draw diagrams showing five different sized regions that you could make with the 20 metres of chicken wire. For each region show the dimensions and the area of the region.



decimals are acceptable as well

Which of your regions, above, has the greatest area? Can you draw a region with a greater area?

The 5×5
 ✓ based on above
 →

This is the greatest area.
 why? $4.5 \times 5.5 = 24.75$
 decimals just less than 25

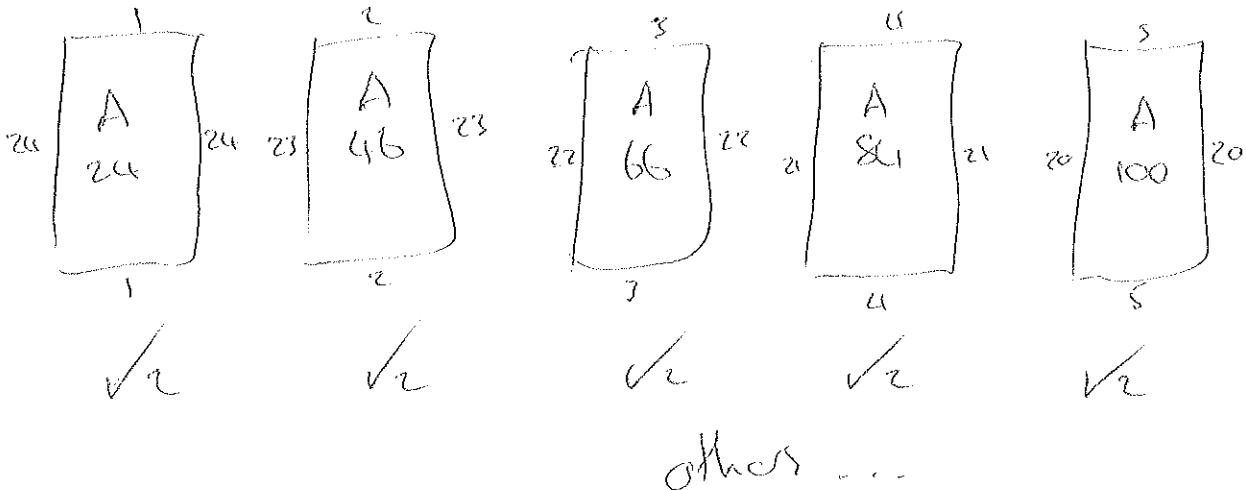
OR ✓ $\sqrt{2}$ justification

No, this instead (if not yet greatest)
 (2) justify

Situation Two (13 marks)

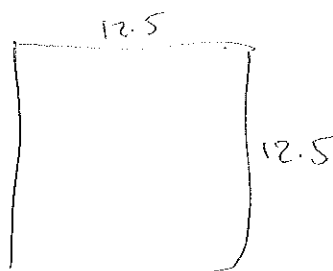
You have 50 metres of chicken wire and you need to enclose a rectangular run.

Draw diagrams showing five different sized regions that you could make with the 50 metres of chicken wire. For each region show the dimensions and the area of the region.



Which of your regions, above, has the greatest area? Can you draw a region with a greater area?

In my answer the 20x5: ✓



$$A = 156.25$$

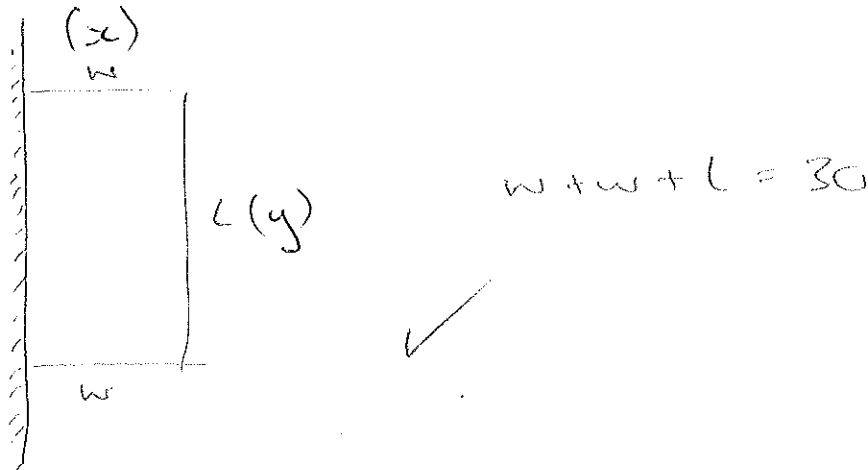
✓ ① For a reasonable response

15

Situation Three (13 marks)

In this new situation you are provided with 30 metres of chicken wire with which to construct 3 sides of this rectangular run, and the fourth side (the length) will be the fence line of the property.

Draw a diagram indicating your understanding of the situation.



Complete the table provided, showing the width (x) of the chicken run, its length (y) and its area (xy).

Width	x	2	3	4	5	6	7	8	9
Length	y	26	24	22	20	18	16	14	12
Area	xy	52	72	88	100	108	112	112	108

Based on the table, what is the maximum area that the chicken run could enclose and what are its dimensions (length and width)?

$$A = 112 \quad \text{for } x = 8 \quad y = 14 \quad \text{and } x = 7 \quad y = 16$$

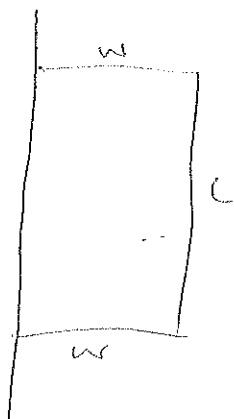
Can you draw a region with a greater area?

$$\text{Yes} \quad x = 7.5 \quad y = 15 \quad A = 112.5$$

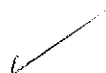
Situation Four (13 marks)

In this new situation you are provided with 50 metres of chicken wire with which to construct 3 sides of this rectangular run, and the fourth side (the length) will be the fence line of the property.

Draw a diagram indicating your understanding of the situation.



$$w + w + L = 50$$



Complete the table provided, showing the width (x) of the chicken run, its length (y) and its area (xy).

Width	x	2	4	6	8	10	12	14	16
Length	y	46	42	38	34	30	26	22	18
Area	xy	92	168	228	272	300	312	308	288

Based on the table, what is the maximum area that the chicken run could enclose and what are its dimensions (length and width)?

$$A = 312, x = 12, y = 26$$

Can you draw a region with a greater area?

Yes, any option is fine. ✓

Actual Max is 312.5, $x = 12.5$, $y = 25$